

[Reading]**1. The Origins of Writing**

It was in Egypt and Mesopotamia (modern-day Iraq) that civilization arose, and it is there that we find the earliest examples of that key feature of civilization, writing. These examples, in the form of inscribed clay tablets that date to shortly before 3000 B.C.E., have been discovered among the archaeological remains of the Sumerians, a gifted people settled in southern Mesopotamia.

The Egyptians were not far behind in developing writing, but we cannot follow the history of their writing in detail because they used a perishable writing material. In ancient times the banks of the Nile were lined with papyrus plants, and from the papyrus reeds the Egyptians made a form of paper; it was excellent in quality but, like any paper, fragile. Mesopotamia's rivers boasted no such useful reeds, but its land did provide good clay, and as a consequence the clay tablet became the standard material. Though clumsy and bulky it has a virtue dear to archaeologists: it is durable. Fire, for example, which is death to papyrus paper or other writing materials such as leather and wood, simply bakes it hard, thereby making it even more durable. So when a conqueror set a Mesopotamian palace ablaze, he helped ensure the survival of any clay tablets in it. Clay, moreover, is cheap, and forming it into tablets is easy, factors that helped the clay tablet become the preferred writing material not only throughout Mesopotamia but far outside it as well, in Syria, Asia Minor, Persia, and even for a while in Crete and Greece. Excavators have unearthed clay tablets in all these lands. In the Near East they remained in use for more than two and a half millennia, and in certain areas they lasted down to the beginning of the common era until finally yielding, once and for all, to more convenient alternatives.

The Sumerians perfected a style of writing suited to clay. This script consists of simple shapes, basically just wedge shapes and lines that could easily be incised in soft clay with a reed or wooden stylus; scholars have dubbed it cuneiform from the wedge-shaped marks (*cunei* in Latin) that are its hallmark. Although the ingredients are merely wedges and lines, there are hundreds of combinations of these basic forms that stand for different sounds or words. Learning these complex signs required long training and much practice; inevitably, literacy was largely limited to a small professional class, the scribes.

The Akkadians conquered the Sumerians around the middle of the third millennium B.C.E., and they took over the various cuneiform signs used for writing Sumerian and gave them sound and word values that fit their own language. ■ The Babylonians and Assyrians did the same, and so did peoples in Syria and Asia Minor. ■ The literature of the Sumerians was treasured throughout the Near East, and long after Sumerian ceased to be spoken, the Babylonians and Assyrians and others kept it alive as a literary language, the way Europeans kept Latin alive after the fall of Rome. ■ For the scribes of these non-Sumerian languages, training was doubly

demanding since they had to know the values of the various cuneiform signs for Sumerian as well as for their own language. ■

The contents of the earliest clay tablets are simple notations of numbers of commodities—animals, jars, baskets, etc. Writing, it would appear, started as a primitive form of bookkeeping. Its use soon widened to document the multitudinous things and acts that are involved in daily life, from simple inventories of commodities to complicated governmental rules and regulations.

Archaeologists frequently find clay tablets in batches. The batches, some of which contain thousands of tablets, consist for the most part of documents of the types just mentioned: bills, deliveries, receipts, inventories, loans, marriage contracts, divorce settlements, court judgments, and so on. These records of factual matters were kept in storage to be available for reference—they were, in effect, files, or, to use the term preferred by specialists in the ancient Near East, archives. Now and then these files include pieces of writing that are of a distinctly different order, writings that do not merely record some matter of fact but involve creative intellectual activity. They range from simple textbook material to literature—and they make an appearance very early, even from the third millennium B C E.

1. The word “key” in the passage is closest in meaning to

- ☐ frequent
- ☐ essential
- ☐ original
- ☐ familiar

2. The word “virtue” in the passage is closest in meaning to

- ☐ price
- ☐ design
- ☐ desirable quality
- ☐ physical characteristic

3. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ In part because of its low cost and ease of use, clay became the preferred writing material throughout Mesopotamia and well beyond it
- ☐ Clay was cheap throughout Mesopotamia, so clay tablets from Mesopotamia became the preferred writing material as far as the Mediterranean.
- ☐ For a while, the day tablet was the preferred writing material in Crete and Greece.
- ☐ Moreover, because day was used as the writing material of choice in Mesopotamia, Syria, Asia Minor, Persia, and the Mediterranean, it was cheap and popular.

4. What can be inferred from paragraph 2 about clay as a writing material?

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- ☐ It had to be baked before it could be written on
 - ☐ Its good points outweighed its bad points.
 - ☐ Its durability was its most important feature for its users.
 - ☐ It was not available in Egypt.

5. In paragraph 2, why does the author discuss the Egyptian use of papyrus as a writing material^

- ☐ To describe the superiority of papyrus over leather and wood as a writing material
- ☐ To explain why writing in Egypt did not develop as quickly as it did Mesopotamia
- ☐ To explain why archaeologists' knowledge of the early history of writing relies mainly on Sumerian cuneiform
- ☐ To explain why the Sumerians preferred clay tablets for writing over papyrus

6. According to paragraph 3, all of the following are true of cuneiform writing EXCEPT:

- ☐ It was composed of very simple shapes
- ☐ It was perfected by the ancient Sumerians.
- ☐ It influenced the choice of material on which it was written.
- ☐ It was understood by very few Sumerians.

7. According to paragraph 4, how did the Akkadians use the Sumerian language?

- ☐ They used Sumerian for speaking but used their own national language for writing.
- ☐ They used the complex cuneiform signs developed by the Babylonians and Assyrians rather than the Sumerian signs.
- ☐ They developed their own cuneiform shapes on clay tablets to replace those used by the Sumerians.
- ☐ They assigned new sound and word values to the signs of Sumerian cuneiform.

8. Paragraph 4 answers all the following questions about Sumerian writing in the period after the Sumerians were conquered EXCEPT:

- ☐ Did Sumerian literature continue to be read?
- ☐ Did Sumerian continue to be spoken?
- ☐ Did scribes compose new texts in Sumerian?
- ☐ Did Sumerian have the same fate as Latin had after the fall of Rome?

9. The word "document" in the passage is closest in meaning to

- ☐ include
- ☐ influence
- ☐ organize
- ☐ record

10. According to paragraph 5, writing was first used for

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- ☐ simple bookkeeping
 - ☐ descriptions of daily events
 - ☐ counting the contents of clay tablets
 - ☐ government reports

11. The phrase “Now and then” in the passage is closest in meaning to

- ☐ always
- ☐ occasionally
- ☐ sooner or later
- ☐ first and last

12. According to paragraph 6, large batches of clay writing tablets were stored because the tablets

- ☐ were being produced quickly and in large quantities
- ☐ did not serve any practical purpose for most Mesopotamians
- ☐ contained information that needed to be available for future reference
- ☐ could not be used again once they had been written on

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

However, the Sumerian language did not entirely disappear.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your answer choices to the spaces where they belong. To remove an answer choice, click on it.

To review the passage, click VIEW TEXT

The earliest examples of writing have been found in Mesopotamia and date to shortly before 3000 B.C.E.



Answer Choices

Writing was invented in the same areas in which civilization began by the ancient civilizations of Mesopotamia, Asia Minor and the Mediterranean.

The development of cuneiform is known because it was written on a long-lasting material and because it was long and widely used throughout the ancient Near East.

Cuneiform tablets generally dealt with business and factual matters, but other topics, including literature, were also recorded and valued.

Writing was developed first by the Sumerians using wedge shaped marks (cuneiform) on clay tablets and then by the Egyptians using papyrus paper.

Scribes using cuneiform in Assyria, Babylon, Syria and Asia Minor had to learn all the languages that used the cuneiform script.

Batches of clay tablets, sometimes with as many as a thousand tablets each, are often found by archaeologists.

2.The Commercial Revolution in Medieval Europe

Beginning in the 1160s, the opening of new silver mines in northern Europe led to the minting and circulation of vast quantities of silver coins. The widespread use of cash greatly increased the volume of international trade. Business procedures changed radically. The individual traveling merchant who alone handled virtually all aspects of exchange evolved into an operation involving three separate types of merchants: the sedentary merchant who ran the "home office," financing and organizing the firm's entire export-import trade; the carriers who transported goods by land and sea; and the company agents resident in cities abroad who, on the advice of the home office, looked after sales and procurements.

Commercial correspondence, unnecessary when one businessperson oversaw everything and made direct bargains with buyers and sellers, multiplied. Regular courier service among commercial cities began. Commercial accounting became more complex when firms had to deal with shareholders, manufacturers, customers, branch offices, employees, and competing firms. Tolls on roads became high enough to finance what has been called a road revolution, involving new surfaces and bridges, new passes through the Alps, and new inns and hospices for travelers. The growth of mutual trust among merchants facilitated the growth of sales on credit

and led to new developments in finance, such as the bill of exchange, a device that made the long, slow, and very dangerous shipment of coins unnecessary.

The ventures of the German Hanseatic League illustrate these advancements. The Hanseatic League was a mercantile association of European towns dating from 1159. The league grew by the end of the fourteenth century to include about 200 cities from Holland to Poland. Across regular, well-defined trade routes along the Baltic and North seas, the ships of league cities carried furs, wax, copper, fish, grain, timber, and wine. These goods were exchanged for finished products, mainly cloth and salt, from western cities. At cities such as Bruges and London, Hanseatic merchants secured special trading concessions, exempting them from all tolls and allowing them to trade at local fairs. Hanseatic merchants established foreign trading centers, the most famous of which was the London Steelyard, a walled community with warehouses, offices, a church, and residential quarters for company representatives. By the late thirteenth century, Hanseatic merchants had developed an important business technique, the business register. Merchants publicly recorded their debts and contracts and received a league guarantee for them. This device proved a decisive factor in the later development of credit and commerce in northern Europe.

These developments added up to what one modern scholar has called "a commercial revolution." In the long run, the commercial revolution of the High Middle Ages (A D 1000-1300) brought about radical change in European society. One remarkable aspect of this change was that the commercial classes constituted a small part of the total population—never more than 10 percent. They exercised an influence far in excess of their numbers. The commercial revolution created a great deal of new wealth, which meant a higher standard of living. The existence of wealth did not escape the attention of kings and other rulers. Wealth could be taxed, and through taxation, kings could create strong and centralized states. In the years to come, alliances with the middle classes were to enable kings to weaken aristocratic interests and build the states that came to be called modern.

The commercial revolution also provided the opportunity for thousands of agricultural workers to improve their social position. The slow but steady transformation of European society from almost completely rural and isolated to relatively more urban constituted the greatest effect of the commercial revolution that began in the eleventh century. Even so, merchants and business people did not run medieval communities, except in central and northern Italy and in the county of Flanders. Most towns remained small. The nobility and churchmen determined the predominant social attitudes, values, and patterns of thought and behavior. The commercial changes of the eleventh through fourteenth centuries did however, lay the economic foundation for the development of urban life and culture.

1. According to paragraph 1, one effect of the increased use of cash was that
O an individual merchant no longer performed all aspects of trading operations

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- ☐ a company's home office declined in importance
 - ☐ merchants no longer had to transport their goods to distant places
 - ☐ the volume of trade declined in areas lacking silver mines

2. The word “radically” , in the passage is closest in meaning to

- ☐ fundamentally
- ☐ quickly
- ☐ unexpectedly
- ☐ gradually

3. The word oversaw" in the passage is closest in meaning to

- ☐ understood
- ☐ included
- ☐ delivered
- ☐ supervised

4. According to paragraph 2, which of the following was NOT an effect of the change in business procedures?

- ☐ An increase in credit sales
- ☐ The use of courier services between cities
- ☐ The adoption of simpler accounting procedures
- ☐ The improvement of roads

5. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ Credit sales and bills of exchange were devices that merchants developed in order to increase their mutual trust.
- ☐ Merchants developed ways to finance their sales without having to rely on slow and dangerous shipments of coins.
- ☐ Greater trust among merchants led to an increase in credit sales and to the use of bills of exchange that made the shipping of coins unnecessary.
- ☐ Merchants began to trust one another when it became too slow and dangerous for a single merchant to ship coins.

6. According to paragraph 3, Hanseatic merchants benefited by all of the following EXCEPT

- ☐ the use of trading centers in distant cities
- ☐ a new system of recording commercial transactions
- ☐ the opening of overland trade routes across northern Europe
- ☐ access to markets in about 200 cities

7. The word "decisive" in the passage is closest in meaning to

- ☐ probable
- ☐ determining
- ☐ helpful
- ☐ limiting

8. Why does the author provide the information in paragraph 4 that the commercial classes never exceeded 10 percent of the population?

- ☐ To argue that the wealth created by the commercial revolution benefited only a small number of people
- ☐ To challenge the view that the commercial classes made up a majority of the population of Europe
- ☐ To suggest a reason that the commercial revolution ended around A. D. 1300
- ☐ To emphasize the point that the commercial revolution was brought about by a small part of the population

9. According to paragraph 4, which of the following was associated with the rise of modern states?

- ☐ Increased wealth for the ruling classes
- ☐ The weakening of the aristocracy
- ☐ The decline of the middle class
- ☐ A reduction in taxes

10. The word "alliances" in the passage is closest in meaning to

- ☐ transactions
- ☐ communications
- ☐ partnerships
- ☐ conflicts

11. According to paragraph 5, the most important result of the commercial revolution was to

- ☐ simplify the organization of European society
- ☐ provide employment to agricultural workers
- ☐ encourage merchants to become community leaders
- ☐ change Europe from a rural to a more urban society

12. Paragraph 5 supports which of the following inferences about the commercial revolution between ad 1000 and 1300?

- ☐ It had very little impact on social attitudes and values.
- ☐ It brought about major political changes throughout Europe.
- ☐ It lessened the influence of the church.

☐ It increased the population of small towns.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

While It originated in the German city of Liibeck, it began to expand in 1241 when Liibeck entered into a mutual protection treaty with the city of Hamburg.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your answer choices to the spaces where they belong To remove an answer choice, click on it

To review the passage, click [VIEW TEXT](#)

During the High Middle Ages (A.D. 1000-1300), Europe underwent a commercial revolution.

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Answer Choices

Merchants adopted new accounting and trading procedures to make long-distance trading more efficient.

The faster transportation made possible by improved roads expanded the variety of goods that could be brought to European towns from far away.

The increasing importance of commercial trade led to a decline in the influence of traditional sources of power, such as kings and church leaders.

The mining of silver improved the security of commercial transactions by allowing coins to replace credit and bills of exchange as the means of exchange.

The Hanseatic League was an association of European towns that obtained shipping, trading, and financial benefits for its members.

European society became increasingly urban, with better living conditions and a stronger centralized government.

3. Ecosystem Diversity and Stability

Conservation biologists have long been concerned that species extinction could have significant consequences for the stability of entire ecosystems—groups of interacting organisms and the physical environment that they inhabit. An ecosystem could survive the loss of some species, but if enough species were lost, the ecosystem would be severely degraded. In fact, it is possible that the loss of a single important species could start a cascade of extinctions that might dramatically change an entire ecosystem. A good illustration of this occurred after sea otters were eliminated from some Pacific kelp (seaweed) bed ecosystems: the kelp beds were practically obliterated too because in the absence of sea otter predation, sea urchin populations exploded and consumed most of the kelp and other macroalgae.

It is usually claimed that species-rich ecosystems tend to be more stable than species-poor ecosystems. Three mechanisms by which higher diversity increases ecosystem stability have been proposed. First, if there are more species in an ecosystem, then its food web will be more complex, with greater redundancy among species in terms of their nutritional roles. In other words, in a rich system if a species is lost, there is a good chance that other species will take over its function as prey, predator, producer, decomposer, or whatever role it played. Second, diverse ecosystems may be less likely to be invaded by new species, notably exotics (foreign species living outside their native range), that would disrupt the ecosystem's structure and function. Third, in a species-rich ecosystem, diseases may spread more slowly because most species will be relatively less abundant, thus increasing the average distance between individuals of the same species and hampering disease transmission among individuals.

Scientific evidence to illuminate these ideas has been slow in coming, and many shadows remain. ■ One of the first studies to provide data supporting a relationship between diversity and stability examined how grassland plants responded to a drought. ■ Researchers D. Tilman and J. A. Downing used the ratio of above-ground biomass in 1988 (after two years of drought) to that in 1986 (predrought) in 207 plots in a grassland field in the Cedar Creek Natural History Area in Minnesota as an index of ecosystem response to disruption by drought. ■ In an experiment that began in 1982, they compared these values with the number of plant species in each plot and discovered that the plots with a greater number of plant species experienced a less dramatic reduction in biomass. ■ Plots with more than ten species had about half as much biomass in 1988 as in 1986, whereas those with fewer than five species only produced roughly one-eighth as much biomass after the two-year drought. Apparently, species-rich plots were likely to contain some drought-resistant plant species that grew better in drought years, compensating for the poor growth of less-tolerant species.

To put this result in more general terms, a species-rich ecosystem may be more

stable because it is more likely to have species with a wide array of responses to variable conditions such as droughts. Furthermore, a species-rich ecosystem is more likely to have species with similar ecological functions, so that if a species is lost from an ecosystem, another species, probably a competitor, is likely to flourish and occupy its functional role. Both of these, variability in responses and functional redundancy, could be thought of as insurance against disturbances.

The Minnesota grassland research has been widely accepted as strong evidence for the diversity- stability theory; however, its findings have been questioned, and similar studies on other ecosystems have not always found a positive relationship between diversity and stability. Clearly, this is a complex issue that requires further field research with a broad spectrum of ecosystems and species: grassland plants and computer models will only take us so far. In the end, despite insightful attempts to detect some general patterns, we may find it very difficult to reduce this topic to a simple, universal truth.

1. The word "significant" in the passage is closest in meaning to
 - ☐ direct
 - ☐ important
 - ☐ long-term
 - ☐ surprising

2. According to paragraph 1, why has the extinction of species been a concern for conservation biologists?
 - ☐ When ecosystems lose just one species, they undergo permanent change.
 - ☐ The extinction of a particular predator species could cause an overpopulation of certain prey species.
 - ☐ The loss of one or more species could cause the decline of a whole ecosystem.
 - ☐ The extinction of a single species is evidence that plant-food sources are in danger of disappearing.

3. According to paragraph 1, what was the result of the removal of Pacific sea otters?
 - ☐ The kelp and sea urchins were destroyed by new predators.
 - ☐ The uncontrolled population of sea urchins ate most of the kelp plants.
 - ☐ Without sea otters, the kelp beds soon became overgrown.
 - ☐ Macroalgae remained as the primary population in the ecosystem.

4. The word "redundancy" in the passage is closest in meaning to
 - ☐ duplication
 - ☐ variety
 - ☐ requirements
 - ☐ flexibility

5. What is the function of paragraph 2 in the passage?

- ☐ To present a hypothesis about ecosystem diversity and some reasons why it might be true
- ☐ To give examples of types of ecosystems that have the greatest diversity
- ☐ To contradict a previous belief about the stability of species-rich ecosystems
- ☐ To contrast species-rich and species-poor ecosystems

6. According to paragraph 2, which of the following increases the stability of an ecosystem?

- ☐ Species in which producers outnumber predators
- ☐ New or exotic species that increase ecosystem diversity
- ☐ Heavily populated species that are free of disease
- ☐ Species that are diverse but have similar nutritional roles

7. Which of the sentences below best expresses the essential information in the highlighted sentence in the passage? Incorrect choices change the meaning in important ways or leave out essential information.

- ☐ In any ecosystem, as the number of individuals in the same species increases, the rate of disease transmission slows down.
- ☐ Ecosystems that have a small number of different species tend to be disease-free, because the species' habitats are at a safe distance from each other.
- ☐ In ecosystems with many species, diseases spread more slowly because there are fewer individuals in a species and, as a result, the individuals are more widely scattered.
- ☐ The average distance between individuals in a species-rich ecosystem increases, so diseases are prevented from being communicated between species.

8. The phrase "compensating for" in the passage is closest in meaning to

- ☐ working against
- ☐ leaving out
- ☐ making up for
- ☐ spreading over

9. What is the main importance of the study discussed in paragraph 3?

- ☐ It examines the response of certain grassland plants to a drought.
- ☐ It contains an index of plants that survived well in times of drought.
- ☐ It provides scientific evidence that diversity helps to make ecosystems stable.
- ☐ It shows that ecosystems contain both resistant species and less tolerant ones.

10. Select the TWO answer choices that, according to paragraph 4, are conclusions that can be drawn from the study by Tilman and Downing. To receive credit you must select TWO answer choices.

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- ☐ A diverse ecosystem will have species that respond differently to a variety of conditions.
 - ☐ Species within a species-rich ecosystem are more likely to have competitors.
 - ☐ An ecosystem is more likely to develop diverse and stable species when it is exposed to extreme conditions.
 - ☐ Species with similar ecological functions will perform the function of a lost species.

11. The word “detect” in the passage is closest in meaning to

- ☐ repeat
- ☐ alter
- ☐ find
- ☐ emphasize

12. According to paragraph 5, which of the following is true about Tilman and Downing’s findings?

- ☐ General patterns of diversity and stability have been established as a result of the findings.
- ☐ Questions about the findings have been refuted by computer models.
- ☐ The findings have been tested in a broad spectrum of ecosystems with similar results.
- ☐ The findings are not sufficient to prove a definite link between diversity and stability in ecosystems.

13. Look at the four squares [■] that indicate where the following sentence could be added to the passage.

It seems clear that there is room for a great deal more research, although some work has been done.

Where would the sentence best fit? Click on a square [■] to add the sentence to the passage.

14. Directions: An introductory sentence for a brief summary of the passage is provided below. Complete the summary by selecting the THREE answer choices that express the most important ideas in the passage. Some sentences do not belong in the summary because they express ideas that are not presented in the passage or are minor ideas in the passage. This question is worth 2 points.

Drag your answer choices to the spaces where they belong To remove an answer choice, click on it

To review the passage, click VIEW TEXT

Biologists have long been worried about the possible effect of the extinction of species on whole ecosystems.

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Answer Choices

Conservation biology studies indicate that the loss of a single important species may bring temporary change to an ecosystem but it seldom results in lasting damage.

Ecosystems having species with similar functions but different responses to adverse conditions can survive environmental disturbances.

The Minnesota grassland study by Tilman and Downing presented evidence that the greater the diversity of species in an ecosystem, the more stable the ecosystem.

The absence of sea otter predation in a Pacific kelp bed ecosystem dramatically changed the entire ecosystem by stabilizing the total kelp population.

The findings of the Minnesota grassland study by Tilman and Downing indicated an equal number of drought-resistant and drought-tolerant plant species in species-rich plots.

More research is needed on the relationship between species diversity and ecosystem stability, though a simple explanation is unlikely.

[Listening]

Conversation 1

1. Why does the student go to see the woman?

☐ To ask about events that the music house sponsors

☐ To find out which of the common interest houses have rooms available

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- ☐ To find out if it would be possible for him to live in the music house
 - ☐ To check on the status of his application to move into the music house

2. According to the woman, why might some people not want to live in the music house?

- ☐ is rarely quiet.
- ☐ is not conveniently located.
- ☐ All of the residents are required to participate in house activities.
- ☐ All of the residents must be enrolled in a music class.

3. What does the woman imply when she mentions the jazz festival?

- ☐ It was free for residents of the music house.
- ☐ It was held at the music house.
- ☐ Music house residents were encouraged to perform at it.
- ☐ Music house residents were involved in organizing it.

4. Why does the woman mention the acceptance rate of applicants for the music house?

- ☐ To the student that his application might not be approved
- ☐ To suggest that the music house is not a popular place to live
- ☐ To convince the student that his chances are better if he applies in person
- ☐ To emphasize the importance of turning in the application form early

5. What information does the student need to include in his application? (Click on 2 answers)

- ☐ Some ways he might contribute to the music house community
- ☐ Information about his experience as a musician
- ☐ Reasons why he wants to live in the music house
- ☐ A recommendation from a current resident of the music house

Lecture 1

1. What is the lecture mainly about?

- ☐ Various methods that ants use to locate food
- ☐ A collective behavior common to humans and animals
- ☐ A type of animal behavior and its application by humans
- ☐ Strategies that flocks of birds use to stay in formation

2. According to the professor, what behavior plays an important role in the way ants obtain food?

- ☐ Ants usually take a different path when they return to their nest.
- ☐ Ants leave chemical trails when they are outside the nest.

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- ☐ Small groups of ants search in different locations.
 - ☐ Ants leave pieces of food along the path as markers.

3. What are two principles of swarm intelligence based on the ant example?
(Click on 2 answers.)

- ☐ Individuals are aware of the group goal.
- ☐ Individuals act on information in their local environment.
- ☐ Individuals follow a leader's guidance.
- ☐ Individuals instinctively follow a set of rules.

4. According to the professor, what path is followed by both telephone calls on a network and ants seeking food?

- ☐ The path with the least amount of activity
- ☐ The most crowded path
- ☐ The path that is most reinforced
- ☐ The path that has intermediate stopping points

5. Why does the professor mention movies?

- ☐ To identify movie scenes with computer-simulated flocks of birds
- ☐ To identify a good source of information about swarm intelligence
- ☐ To emphasize how difficult it still is to simulate bird flight
- ☐ To explain that some special effects in movies are based on swarm intelligence

6. What is the professor's attitude about attempts to create computer-generated crowds of people?

- ☐ She believes that the rules of birds' flocking behavior do not apply to group behavior in humans.
- ☐ She thinks that crowd scenes could be improved by using the behavior of ant colonies as a model.
- ☐ She is surprised by how realistic the computer-generated crowds are.
- ☐ She is impressed that computer graphics can create such a wide range of emotions.

Lecture 2

1. What is the lecture mainly about?

- ☐ Techniques for creating lifelike portraits
- ☐ Theories about a preference in portrait painting
- ☐ Reasons for a decline in the popularity of portrait painting
- ☐ Ways researchers determine whether a painting is authentic

2. According to the professor, what change occurred in portrait paintings over time?

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- ☐ Portrait artists became more skilled at painting profiles.
 - ☐ Portrait artists began painting a greater variety of subjects.
 - ☐ The percentage of left profiles decreased.
 - ☐ A left profile became a symbol of the subject's social status.

3. What point does the professor emphasize by his discussion of the lithograph *The Potato Eaters*?

- ☐ Profiles are more difficult to create than head-on portraits.
- ☐ Some artists consider the direction of the profile a key element in a portrait.
- ☐ Most group portraits include both left and right profiles.
- ☐ The direction of a profile is usually determined by the characteristics of the subject.

4. According to the professor, what is the problem with the "right-handed" hypothesis?

- ☐ Left-handed artists are equally comfortable painting left and right profiles.
- ☐ Right hands tend to cover up facial features when painting left profiles.
- ☐ Statistics show that right-handed artists tend to paint right profiles.
- ☐ Left hands are less likely to cast shadows when painting right profiles.

5. What does the professor imply about the "studio organization" hypothesis?

- ☐ It has been disproved by recent studies.
- ☐ It accounts for an important change in portrait painting.
- ☐ It is based on a study of a relatively small number of portraits.
- ☐ It was first proposed by the owner of a modern-day portrait studio.

6. Why does the student say this:

- ☐ To explain why he prefers to paint left profiles
- ☐ To point out a difference between adult portraits and baby portraits
- ☐ To point out a fact that contradicts the "parental imprinting" hypothesis
- ☐ To provide support for the "parental imprinting" hypothesis

Conversation 2

1. What is the conversation mainly about?

- ☐ Using new technologies to preserve old newspapers
- ☐ Using old newspapers to conduct historical research
- ☐ The rise of American journalism in the eighteenth century
- ☐ Press coverage of the French Revolution of 1789

2. What gave the student inspiration for the topic of her term paper?(Click on 2 answers.)

-
- ☐ A recent visit to the library's microfilm collection
 - ☐ A long-standing interest in the history of France
 - ☐ Seeing what an eighteenth-century newspaper looked like
 - ☐ Reading a translation of French historical accounts

3. According to the professor, what should the student ask the librarians?

- ☐ Which eighteenth-century newspapers covered events in France
- ☐ If she can request microfilmed newspapers from another library
- ☐ If the old newspapers she wants to read are available online yet
- ☐ Whether the library has any original copies of eighteenth-century newspapers

4. What will the student probably include in her term paper?

- ☐ Newspaper coverage of the French National Constituent Assembly
- ☐ Newspaper coverage of the storming of the Bastille prison
- ☐ Ways in which the French Revolution contributed to the development of democratic ideals
- ☐ How the reporting of American journalists differed from that of French journalists

5. What can be inferred about the professor when he discusses a paper presented at a history conference?

- ☐ He is grateful that he saved the paper because it might help the student.
- ☐ He worries that the student will overgeneralize American attitudes based on the content of newspapers
- ☐ He is excited to provide a model that the student can use to organize her term paper.
- ☐ He hopes that the student will consider interviewing the author of the paper.

Lecture 3

1. What is the lecture mainly about?

- ☐ The economic factors that made lapis lazuli expensive
- ☐ The types of paintings in which the color blue is popular
- ☐ Early processes for making blue pigments from stones
- ☐ Difficulties using the color blue in early paintings

2. What was Gainsborough's goal when he painted *The Blue Boy*?

- ☐ To find an acceptable alternative to ultramarine pigment
- ☐ To demonstrate that blue should be used only in certain paintings
- ☐ To contradict a common belief about the use of blue in a painting
- ☐ To protest the high costs of painting with most blue pigments

3. What does the professor imply about smalt as a substitute for lapis lazuli?

-
- ☐ It eventually became as expensive as lapis lazuli.
 - ☐ It was used frequently throughout the nineteenth century.
 - ☐ It was not of an acceptable quality for many artists.
 - ☐ It was seen as a better substitute for lapis lazuli than azurite was.

4. What two points does the professor make about the process of turning lapis lazuli into ultramarine? (Click on 2 answers.)

- ☐ It took a lot of time.
- ☐ It required expensive tools.
- ☐ It did not produce much pigment.
- ☐ It was perfected by the French.

5. Why does the professor mention the French government?

- ☐ To indicate who sponsored the digging of additional lapis lazuli mines
- ☐ To emphasize the importance of developing an affordable blue pigment
- ☐ To point out that artists were not permitted to use certain stones to make pigments
- ☐ To question the government's use of public funds to support the arts

6. What does the professor imply when he says this:

- ☐ He is not convinced the Egyptians made the first synthetic pigment
- ☐ He is impressed by the Egyptians' accomplishment.
- ☐ The process the Egyptians used is now widely known.
- ☐ The Egyptian pigment was of lower quality than today's pigments.

Lecture 4

1. What does the professor mainly discuss? (Click on 2 answers.)

- ☐ Reasons for fluctuations in the price of copper
- ☐ Some important attributes of copper
- ☐ The production of coins from copper and zinc
- ☐ The possible future of the United States penny

2. What arguments in favor of keeping the penny in circulation does the professor emphasize? (Click on 2 answers.)

- ☐ Some people are emotionally attached to pennies.
- ☐ Pennies cost very little to produce.
- ☐ The price of consumer goods could rise.
- ☐ The copper industry's profits depend on the production of pennies.

3. What does the professor say about the negative seigniorage of the nickel?

- ☐ The United States government is looking for ways to reduce it.

-
- ☐ It is significantly greater than that of the penny.
 - ☐ It is less of a problem than some people believe it to be.
 - ☐ Merchants benefit from it more than consumers do.

4. Why does the professor mention the trumpet and trombone?

- ☐ To compare the sound-generating properties of copper with those of zinc
- ☐ To exemplify the benefits of mixing copper with other metals
- ☐ To point out that copper can be shaped into a variety of forms
- ☐ To point out that objects containing copper can shine like gold

5. What does the professor imply about the green patina that sometimes appears on copper?

- ☐ It is difficult to remove by conventional means.
- ☐ It adds to the beauty of objects made of copper.
- ☐ It serves a useful function.
- ☐ It reduces the conductivity of copper.

6. Why does the professor say this:

- ☐ He wants the woman to realize her own mistake.
- ☐ He wants the woman to support her point with precise numbers.
- ☐ He realizes he neglected to mention an important detail.
- ☐ He shares the woman's concern about the copper industry.

[Speaking]

1. Talk about one way in which you have benefited from using the Internet. Use specific examples and details in your response.

2. Do you agree or disagree with the following statement?

**People today have healthier lifestyles than people did 100 years ago.
Use specific examples and details to support your opinion.**

3. Reading Time

No More Posters Outside the Student Center

Traditionally, students have been permitted to put up posters advertising events, clubs, and activities on the concrete wall outside the main entrance to the student center. Beginning next semester, however, students will no longer be allowed to attach anything to the front of the building. According to a university official, the new policy is part of an effort to improve the campus's appearance. "The posters on the wall make the building unattractive," he said, "so they have to come down." He added that posters can instead be displayed in the dining hall: "Students

can use the bulletin board in the dining hall for posters, so the policy change should not be a problem.

The woman expresses her opinion about the new policy. Briefly describe the policy. Then state the woman's opinion about the policy, and explain the reasons she gives for holding that opinion.

4. Reading Time

Warning Coloration

Many animals have defense mechanisms that help protect them from predators. In some cases, these animals have distinct coloring that signals predators of the presence of such defenses. This type of coloring is called warning coloration. If a predator does not recognize the meaning of this coloration and attacks, it may suffer significant discomfort or injury when its would-be prey employs its defense mechanism. As a result, the predator learns to associate the warning coloration with negative consequences, and will from that point forward avoid attacking animals that have that coloration.

Explain how the professor's example from the lecture illustrates warning coloration.

5.

Briefly summarize the problem the speakers are discussing. Then state which of the two solutions from the conversation you would recommend. Explain the reasons for your recommendation.

6.

Using points and examples from the lecture, give two explanations for why we may remember some things better than others.

[Writing]**1. (Reading)**

In the United States, medical information about patients traditionally has been recorded and stored on paper forms. However, there are efforts to persuade doctors to adopt electronic medical record systems in which information about patients is stored in electronic databases rather than on paper. It is argued that storing patients' medical records in electronic databases has several advantages over traditional paper-based record keeping.

Reducing Costs

First, the use of electronic records can help reduce costs by saving money on storing and transferring medical records. While paper records require a significant amount of storage space, electronic medical records take up virtually no space. Moreover, by having patients' records computerized in databases, doctors can easily access the records from almost anywhere and can easily duplicate and transfer them when necessary. This costs much less than copying, faxing, or transporting paper records from one location to another.

Preventing Errors

Second, electronic medical records are crucial to reducing the chances of medical errors. Illegible handwriting, improper transcription of data, and nonstandard organization of paper records have caused errors that in some cases have had serious consequences for the patients' health. In contrast, electronic records are associated with standardization of forms and legible computer fonts and thus minimize the possibility of human error.

Aiding Research

Third, electronic medical records can greatly aid medical research by making it possible to gather large amounts of data from patient records. It is often impractical, impossible, or prohibitively expensive to manually go through thousands of patients' paper records housed in doctors' offices. However, with the existence of electronic medical records, it would be simple to draw out the needed information from the medical databases because the databases are already formatted for data collection. Once in the electronic system, the records could be accessed from any research location.

Directions: You have 20 minutes to plan and write your response. Your response will be judged on the basis of the quality of your writing and on how well your response presents the points in the lecture and their relationship to the reading passage. Typically, an effective response will be 150 to 225 words.

Question: Summarize the points made in the lecture, being sure to explain how they cast doubt on the specific points made in the reading passage.

2. Directions: Read the question below. You have 30 minutes to plan, write, and revise your essay. Typically, an effective response will contain a minimum of 300 words.

Question: Do you agree or disagree with the following statement?

The opinions of celebrities, such as famous entertainers and athletes, are more important to younger people than they are to older people.

Use specific reasons and examples to support your answer.